

Question 1

1 mark 101

There are 24 different movies Kiandra can download to her computer. Determine the number of ways she can select 15 movies.

Question 2

a) 1 mark b) 1 mark

102
103

Given $\theta = 40^\circ$,

a) convert θ to radians.

b) determine the coterminal angles of θ where $\theta \in \mathbb{R}$.

Question 3

3 marks 104

Peter invests \$560 per month at an annual interest rate of 4.2%, compounded monthly. Determine how many monthly investments he will need to make to obtain at least \$500 000.

Express your answer as a whole number.

Use the formula:

$$FV = \frac{R[(1+i)^n - 1]}{i}$$

where FV = the future value

R = the investment amount each period

$i = \frac{\text{the annual interest rate}}{\text{the number of compounding periods per year}}$

n = the number of investments

Question 4

2 marks 105

Ishmael has 4 dogs, 5 cats, and 3 horses.

If he arranges all of them in a row, determine how many ways they can be arranged if each type of animal must be grouped together.

Note: A calculator is not required for the remaining test questions.

Question 5

3 marks 106

Solve the following equation algebraically over the interval $0 \leq \theta \leq 2\pi$.

$$2 \cos^2 \theta + 9 \cos \theta - 5 = 0$$

Question 6

2 marks 107

Determine which term contains $\frac{1}{x^6}$ in the binomial expansion of $\left(\frac{2}{x^3} + 3x^2\right)^7$.

Question 7

1 mark 108

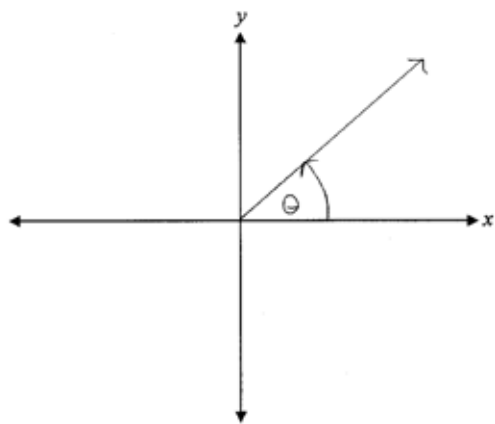
Determine the radius of a circle which has an arc length of 5 cm with a central angle of 3 radians.

Question 8

1 mark 109

Tyler incorrectly sketched the angle $\theta = -\frac{7\pi}{4}$ in standard position.

Describe his error.



Question 9

a) 1 mark b) 3 marks

110
111

Given the identity $\sec \theta + \cos \theta = \frac{2 - \sin^2 \theta}{\cos \theta}$,

a) determine the non-permissible values of θ , over the interval $0 \leq \theta \leq 2\pi$.

b) prove the identity for all permissible values of θ .

Left-Hand Side	Right-Hand Side

Question 10

2 marks 112

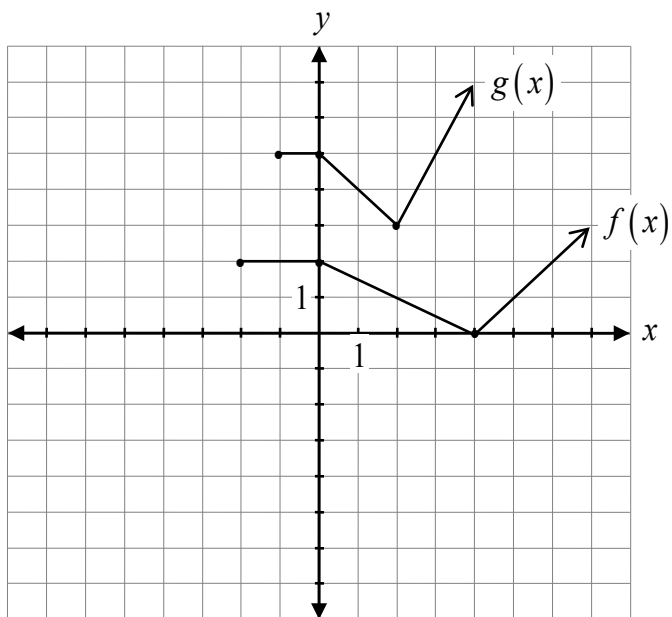
Expand using the laws of logarithms.

$$\log\left(\frac{a}{b^4}\right)$$

Question 11

2 marks 113

State the equation of $g(x)$ in terms of $f(x)$.

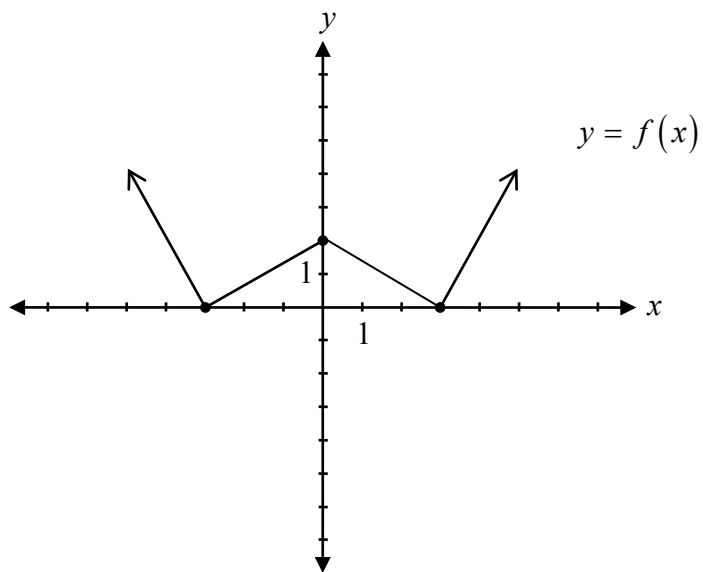


$g(x) =$ _____

Question 12

1 mark 114

Explain why the inverse of the graph of $y = f(x)$ is not a function.



Question 13

2 marks ¹¹⁵

Solve the following equation algebraically:

$$\log(x^2 + 5) - \log(x^2 + 1) = \log 3$$

Question 14

2 marks 116

Describe the transformations used to obtain the graph of the function $y = 5f(x + 1)$ from the graph of $y = f(x)$.

Question 15

3 marks ¹¹⁷

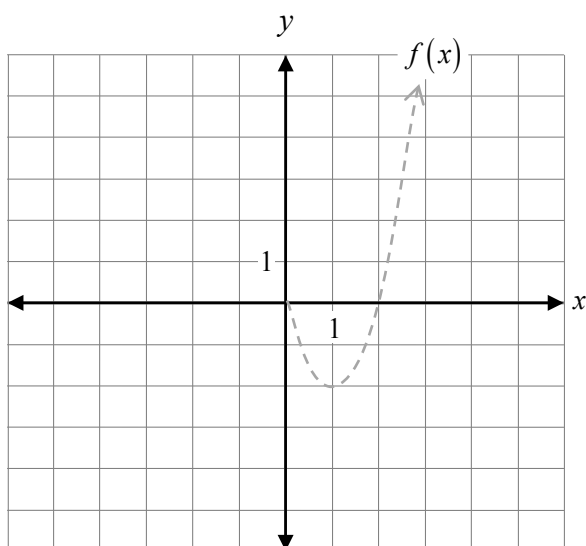
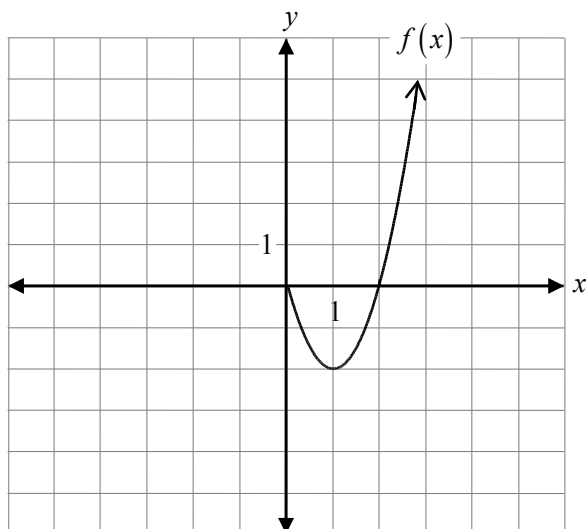
Solve algebraically:

$${}_nC_2 = 3n + 4$$

Question 16

3 marks 118

Given the graph of $f(x)$, sketch the graph of $y = \left| \frac{1}{2} f(x-1) \right|$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 17

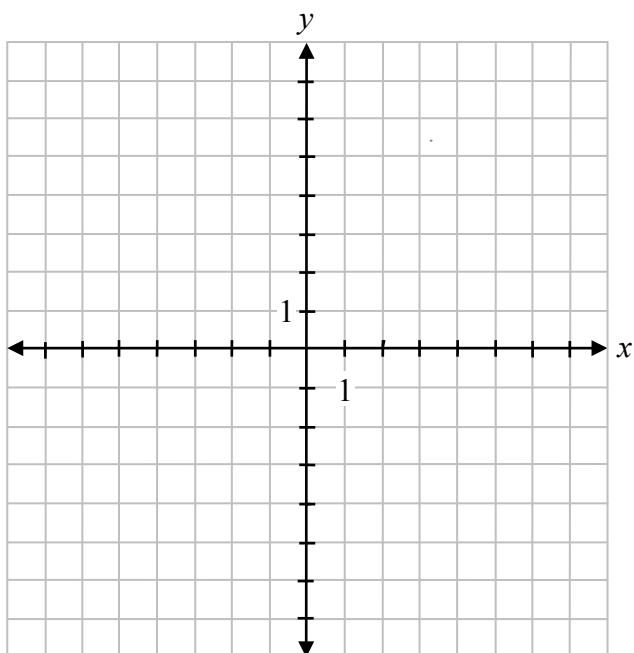
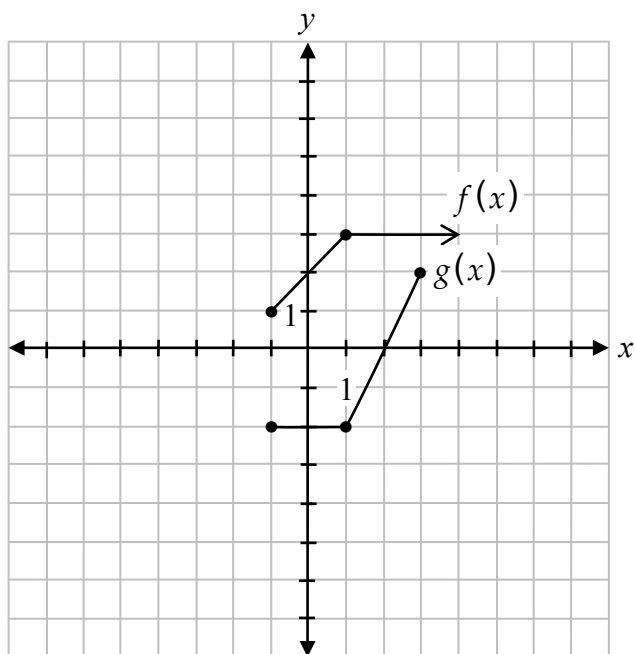
1 mark 119

Explain why $f(x) = (x+2)^3(x-1)^{\frac{1}{2}}$ is not a polynomial function.

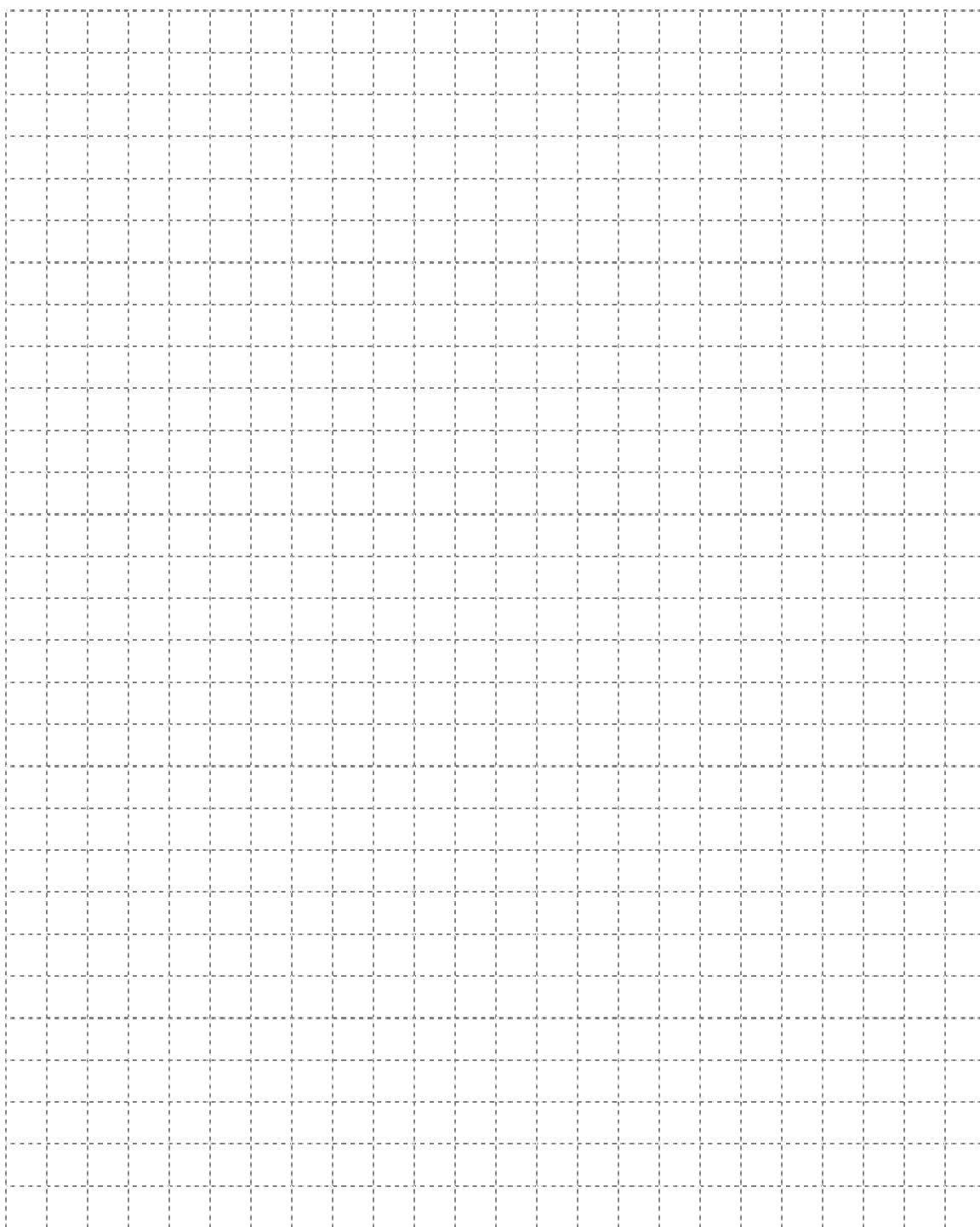
Question 18

2 marks 120

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = (f \circ g)(x)$.



No marks will be awarded for work done on this page.



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